POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6014C | Semester VI | Theory | 4 Credits

# Green Building and Energy Conservation

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

| | |
|---|---|
| Programmes | **Civil Engineering** |
| Contact hours | **60** |
| Teaching scheme | **3:1:0:0** |
| Credits | **4** |
| CIE / ESE | **Theory CIE 40 / Theory ESE 60** |
| Self-learning | **5.5 h/week** |

CURRICULUM INTEGRITY

## Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6014C as Green Building and Energy Conservation in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Delhi’s Sustainable Materials and Green Buildings course, the Swayam course on Energy Efficient Buildings and polytechnic green building curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Energy audits, green building certifications, water-saving designs and material specifications must be based on approved engineering designs, certified assessment tools (GRIHA/LEED), current building codes (ECBC) and competent professional review.

> **Verified source note:** Verified sources support embodied/operational energy, ecological footprint, sustainable materials, indoor air quality, ECBC, LEED, GRIHA, orientation, shading, insulation and passive cooling. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

## What You Are Expected to Learn

**60**
Contact hours

**4**
Credits

**Theory CIE 40**
CIE

**Theory ESE 60**
ESE

## Course introduction

Green Building and Energy Conservation studies the design, construction and operation of buildings that are environmentally responsible and resource-efficient throughout their life cycle. It develops the ability to apply energy-efficient design principles, select sustainable materials, implement water conservation strategies, and understand green building rating systems (GRIHA, LEED) while respecting the technical and regulatory boundaries of sustainable construction.

> **Malayalam support:** ഈ handbook പഠിക്കുമ്പോൾ syllabus topic ആദ്യം വായിക്കുക; പിന്നീട് principle, diagram/procedure, application, common mistake എന്നിവ ക്രമമായി പഠിക്കുക.

## Prerequisite foundation

Building construction, construction materials, environmental studies, elementary physics and basic hydraulics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

**Teaching and assessment scheme**

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE | ESE |
|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 0 | 0 | 5.5 h/week | 60 | 4 | Theory CIE 40 | Theory ESE 60 |

STUDY METHOD

# How to Use This Handbook

### 1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

### 4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

## Objectives and Course Outcomes

### Official course objectives

1. Explain green building fundamentals, benefits and the features of sustainable site selection.
2. Explain energy-efficient design strategies, including passive heating/cooling and building orientation.
3. Explain green building materials, water conservation techniques and indoor environmental quality.
4. Explain energy conservation codes (ECBC), energy auditing and green building rating systems in India.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO എന്നത് exam-ൽ എന്ത് ചെയ്യാൻ കഴിയണം എന്ന് പറയുന്നു. Define, explain, apply എന്നീ action words ശ്രദ്ധിക്കുക.

**Official Course Outcomes**

| CO | Official outcome | Bloom level |
|---|---|---|
| CO1 | Explain the concepts of green buildings and the principles of sustainable site planning. | Understand |
| CO2 | Explain energy-efficient design strategies and passive cooling/heating techniques. | Understand |
| CO3 | Explain the selection of green materials and the implementation of water conservation systems. | Understand |
| CO4 | Apply energy conservation codes and evaluate buildings using green rating systems. | Apply |

**Official CO–PO articulation matrix**

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7–PO11 |
|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO2 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO3 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO4 | 3 | 2 | 2 | 2 | 2 | – | – |

SYLLABUS COVERAGE

## Curriculum Coverage Map

| Module | Title | Official major topics | Hours | CO | Handbook section |
|---|---|---|---|---|---|
| All official major topics mapped to handbook chapters | | | | | |
| 1 | Green Building Fundamentals and Site Planning | Definition and characteristics of green buildings; benefits (environmental, economic, social); sustainable site selection and development; minimizing site disturbance; heat island effect; ecological footprint of buildings. | Approx. 14 hours | CO1 | Module 1 |
| 2 | Energy Efficient Design and Passive Strategies | Building orientation and form; shading devices (internal and external); thermal insulation; building envelope design; passive heating and cooling techniques (ventilation, thermal mass); energy efficient lighting and HVAC. | Approx. 16 hours | CO2 | Module 2 |
| 3 | Green Materials and Water Conservation | Sustainable building materials (low-carbon cement, recycled aggregate, fly ash); embodied energy of materials; water conservation strategies; rainwater harvesting; greywater treatment and reuse; low-flow fixtures. | Approx. 16 hours | CO3 | Module 3 |
| 4 | Energy Codes, Auditing and Rating Systems | Energy Conservation Building Code (ECBC); principles of energy auditing; green building rating systems in India (GRIHA, LEED, IGBC); indoor environmental quality (VOC, ventilation); life cycle cost analysis. | Approx. 14 hours | CO4 | Module 4 |

LEARNING ROADMAP

## How the Modules Connect

### Green Building Fundamentals and Site Planning

Definition and characteristics of green buildings; benefits (environmental, economic, social); sustainable site selection and development; minimizing site disturbance; heat island effect; ecological footprint of buildings.

CO1 · Approx. 14 hours

### Energy Efficient Design and Passive Strategies

Building orientation and form; shading devices (internal and external); thermal insulation; building envelope design; passive heating and cooling techniques (ventilation, thermal mass); energy efficient lighting and HVAC.

CO2 · Approx. 16 hours

### Green Materials and Water Conservation

Sustainable building materials (low-carbon cement, recycled aggregate, fly ash); embodied energy of materials; water conservation strategies; rainwater harvesting; greywater treatment and reuse; low-flow fixtures.

CO3 · Approx. 16 hours

### Energy Codes, Auditing and Rating Systems

Energy Conservation Building Code (ECBC); principles of energy auditing; green building rating systems in India (GRIHA, LEED, IGBC); indoor environmental quality (VOC, ventilation); life cycle cost analysis.

CO4 · Approx. 14 hours

MODULE 1

## Green Building Fundamentals and Site Planning

Definition and characteristics of green buildings; benefits (environmental, economic, social); sustainable site selection and development; minimizing site disturbance; heat island effect; ecological footprint of buildings.

**Approx. 14 hours** **CO1** **Understand · Apply**

### Learning goals

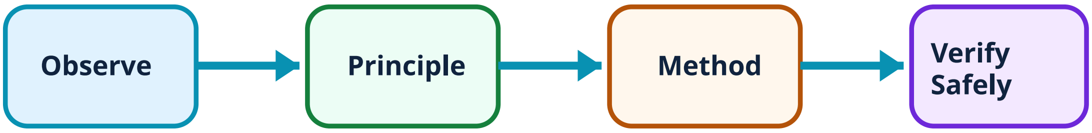
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Green Building Fundamentals and Site Planning: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Green building concepts and benefits–

Green buildings aim to reduce the overall impact of the built environment on human health and the natural environment by efficiently using energy, water and other resources. Benefits include lower operating costs, improved occupant productivity and reduced environmental footprint.

### Study and application method

List four features of a green building and explain how they differ from conventional construction; define 'ecological footprint' in the context of building development.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** List four features of a green building and explain how they differ from conventional construction; define 'ecological footprint' in the context of building development.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Green building is a holistic process. Do not focus on a single feature (e.g., solar panels) without considering the overall resource efficiency and site impact.

## 2. Sustainable site selection and design–

Site planning involves selecting locations with good connectivity, protecting existing vegetation and minimizing soil erosion. Strategies like 'cool roofs' and urban greenery help mitigate the 'heat island effect'—where urban areas are significantly warmer than their surroundings.

### Study and application method

Describe three strategies for sustainable site development; explain the causes and mitigation measures for the 'urban heat island effect'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe three strategies for sustainable site development; explain the causes and mitigation measures for the 'urban heat island effect'.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Site development must comply with local zoning and environmental regulations. Do not assume that green features exempt a project from authorised land-use requirements.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

## Energy Efficient Design and Passive Strategies

Building orientation and form; shading devices (internal and external); thermal insulation; building envelope design; passive heating and cooling techniques (ventilation, thermal mass); energy efficient lighting and HVAC.

Approx. 16 hours | CO2 | Understand · Apply

### Learning goals

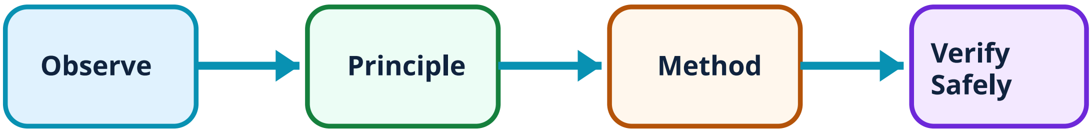
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Energy Efficient Design and Passive Strategies: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Orientation and shading–

Proper orientation maximizes natural light and minimizes unwanted solar heat gain. Shading devices (chajjas, fins, blinds) are designed based on solar-path analysis to protect the building envelope during peak heat hours while allowing winter sun.

### Study and application method

Sketch the ideal orientation for a building in a hot-dry climate and identify two types of external shading devices.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Sketch the ideal orientation for a building in a hot-dry climate and identify two types of external shading devices.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Shading design is climate-specific. Use authorised solar-path data and local climatic records for all energy-efficient design calculations.

## 2. Passive cooling and heating–

Passive strategies use natural processes to maintain comfort. Techniques include cross-ventilation, stack effect, evaporative cooling and the use of high-thermal-mass materials. These reduce the reliance on active, energy-intensive mechanical systems.

### Study and application method

Explain the working principle of 'cross-ventilation' and 'thermal mass' in maintaining indoor thermal comfort.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Explain the working principle of 'cross-ventilation' and 'thermal mass' in maintaining indoor thermal comfort.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Passive strategies require careful integration with building use and occupant behavior. Do not assume passive cooling will meet all comfort needs in extreme climates without authorised simulation.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

## Green Materials and Water Conservation

Sustainable building materials (low-carbon cement, recycled aggregate, fly ash); embodied energy of materials; water conservation strategies; rainwater harvesting; greywater treatment and reuse; low-flow fixtures.

Approx. 16 hours | CO3 | Understand · Apply

### Learning goals

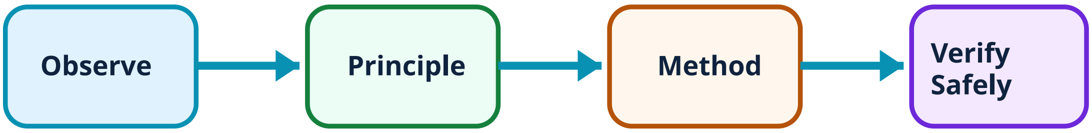
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Green Materials and Water Conservation: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Sustainable materials and embodied energy–

Green materials have low 'embodied energy'—the energy required for extraction, processing and transport. Using industrial by-products (fly ash, GGBS) and recycled materials reduces the environmental impact of construction.

### Study and application method

Compare the embodied energy of three common building materials (e.g., brick, concrete, steel); list the advantages of using recycled aggregates.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare the embodied energy of three common building materials (e.g., brick, concrete, steel); list the advantages of using recycled aggregates.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Sustainable materials must meet the same structural and durability standards as conventional ones. Do not use unverified materials for safety-critical components.

## 2. Water conservation and management–

Buildings conserve water through demand reduction (low-flow fixtures), rainwater harvesting for recharge or use, and onsite treatment and reuse of greywater for landscaping and flushing, significantly reducing the load on municipal supplies.

### Study and application method

Draw a conceptual layout for a rainwater harvesting system for a residential building; explain the difference between 'black water' and 'grey water'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Draw a conceptual layout for a rainwater harvesting system for a residential building; explain the difference between 'black water' and 'grey water'.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Water reuse systems must prevent cross-contamination with potable supplies. All plumbing for non-potable water must be clearly marked and comply with authorised health codes.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

## Energy Codes, Auditing and Rating Systems

Energy Conservation Building Code (ECBC); principles of energy auditing; green building rating systems in India (GRIHA, LEED, IGBC); indoor environmental quality (VOC, ventilation); life cycle cost analysis.

Approx. 14 hours | CO4 | Understand · Apply

### Learning goals

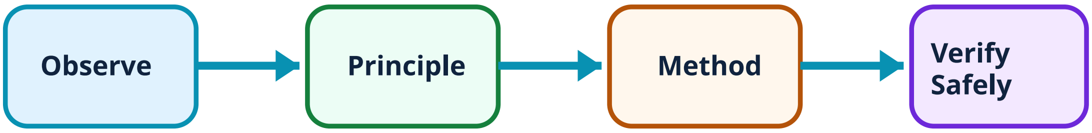
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Energy Codes, Auditing and Rating Systems: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Energy codes and auditing–

The ECBC sets minimum energy performance standards for commercial buildings. Energy auditing is a systematic process of identifying energy-saving opportunities in existing buildings through measurement and analysis of energy use patterns.

## Study and application method

Identify the major components of the Energy Conservation Building Code (ECBC); describe the basic steps in conducting a preliminary energy audit.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

## Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify the major components of the Energy Conservation Building Code (ECBC); describe the basic steps in conducting a preliminary energy audit.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Energy auditing requires specialized equipment and expertise. All audits must be conducted by certified energy auditors according to authorised protocols.

## 2. Green rating systems and IEQ–

Rating systems like GRIHA (National) and LEED/IGBC (International) provide a framework for assessing and certifying building sustainability. Indoor Environmental Quality (IEQ) focuses on occupant comfort through low-VOC materials and adequate ventilation.

### Study and application method

Compare the GRIHA and LEED rating systems based on their assessment criteria; explain the importance of 'Indoor Air Quality' (IAQ) for occupant health.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare the GRIHA and LEED rating systems based on their assessment criteria; explain the importance of 'Indoor Air Quality' (IAQ) for occupant health.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Rating systems are tools for assessment, not substitutes for building codes. Certification must be performed by authorised third-party agencies based on verified project data.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

## Rapid Recall Bank

### Recall 1

State symbols and SI units before substitution.

### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

## Diagram Library for Rapid Revision

### Module 1: Green Building Fundamentals and Site Planning

Use the concept flow to revise observation, principle, method and verification.

### Module 2: Energy Efficient Design and Passive Strategies

Use the concept flow to revise observation, principle, method and verification.

### Module 3: Green Materials and Water Conservation

Use the concept flow to revise observation, principle, method and verification.

### Module 4: Energy Codes, Auditing and Rating Systems

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Compare three green building materials with their conventional alternatives in a conceptual table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Sketch the solar shading for a south-facing window in your region for summer and winter.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Calculate the potential rainwater harvest for a building conceptually given its roof area and local rainfall.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Identify the key criteria for a 'GRIHA' or 'LEED' rating for a commercial building.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Prepare a checklist for improving energy efficiency in an existing residential building.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

## Assessment Methodology and Examination Pattern

**Official assessment structure**

| Component | Official mark / conversion | Student evidence |
|---|---|---|
| Attendance | 5 marks | End-of-semester attendance component. |
| Self-learning activities | 15 marks | Module-linked activities. |
| Series examinations | 20 + 20 converted | Two 50-mark series examinations. |
| CIE total | Theory CIE 40 | Continuous internal evaluation. |
| Written ESE | Theory ESE 60 | 2.5 hours; official Part A / Part B / Part C pattern. |

### Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

### Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

## Module-wise Questions with Answers

## Module 1 — Green Building Fundamentals and Site Planning

**Explain Green building concepts and benefits and state one correct method, application or precaution.–**

Green buildings aim to reduce the overall impact of the built environment on human health and the natural environment by efficiently using energy, water and other resources. Benefits include lower operating costs, improved occupant productivity and reduced environmental footprint.

**Answer framework:** List four features of a green building and explain how they differ from conventional construction; define 'ecological footprint' in the context of building development.

**Important point:** Green building is a holistic process. Do not focus on a single feature (e.g., solar panels) without considering the overall resource efficiency and site impact.

**Explain Sustainable site selection and design and state one correct method, application or precaution.–**

Site planning involves selecting locations with good connectivity, protecting existing vegetation and minimizing soil erosion. Strategies like 'cool roofs' and urban greenery help mitigate the 'heat island effect'—where urban areas are significantly warmer than their surroundings.

**Answer framework:** Describe three strategies for sustainable site development; explain the causes and mitigation measures for the 'urban heat island effect'.

**Important point:** Site development must comply with local zoning and environmental regulations. Do not assume that green features exempt a project from authorised land-use requirements.

## Module 2 — Energy Efficient Design and Passive Strategies

**Explain Orientation and shading and state one correct method, application or precaution.–**

Proper orientation maximizes natural light and minimizes unwanted solar heat gain. Shading devices (chajjas, fins, blinds) are designed based on solar-path analysis to protect the building envelope during peak heat hours while allowing winter sun.

**Answer framework:** Sketch the ideal orientation for a building in a hot-dry climate and identify two types of external shading devices.

**Important point:** Shading design is climate-specific. Use authorised solar-path data and local climatic records for all energy-efficient design calculations.

**Explain Passive cooling and heating and state one correct method, application or precaution.–**

Passive strategies use natural processes to maintain comfort. Techniques include cross-ventilation, stack effect, evaporative cooling and the use of high-thermal-mass materials. These reduce the reliance on active, energy-intensive mechanical systems.

**Answer framework:** Explain the working principle of 'cross-ventilation' and 'thermal mass' in maintaining indoor thermal comfort.

**Important point:** Passive strategies require careful integration with building use and occupant behavior. Do not assume passive cooling will meet all comfort needs in extreme climates without authorised simulation.

## Module 3 — Green Materials and Water Conservation

**Explain Sustainable materials and embodied energy and state one correct method, application or precaution.–**

Green materials have low 'embodied energy'—the energy required for extraction, processing and transport. Using industrial by-products (fly ash, GGBS) and recycled materials reduces the environmental impact of construction.

**Answer framework:** Compare the embodied energy of three common building materials (e.g., brick, concrete, steel); list the advantages of using recycled aggregates.

**Important point:** Sustainable materials must meet the same structural and durability standards as conventional ones. Do not use unverified materials for safety-critical components.

**Explain Water conservation and management and state one correct method, application or precaution.–**

Buildings conserve water through demand reduction (low-flow fixtures), rainwater harvesting for recharge or use, and onsite treatment and reuse of greywater for landscaping and flushing, significantly reducing the load on municipal supplies.

**Answer framework:** Draw a conceptual layout for a rainwater harvesting system for a residential building; explain the difference between 'black water' and 'grey water'.

**Important point:** Water reuse systems must prevent cross-contamination with potable supplies. All plumbing for non-potable water must be clearly marked and comply with authorised health codes.

## Module 4 — Energy Codes, Auditing and Rating Systems

**Explain Energy codes and auditing and state one correct method, application or precaution.–**

The ECBC sets minimum energy performance standards for commercial buildings. Energy auditing is a systematic process of identifying energy-saving opportunities in existing buildings through measurement and analysis of energy use patterns.

**Answer framework:** Identify the major components of the Energy Conservation Building Code (ECBC); describe the basic steps in conducting a preliminary energy audit.

**Important point:** Energy auditing requires specialized equipment and expertise. All audits must be conducted by certified energy auditors according to authorised protocols.

**Explain Green rating systems and IEQ and state one correct method, application or precaution.–**

Rating systems like GRIHA (National) and LEED/IGBC (International) provide a framework for assessing and certifying building sustainability. Indoor Environmental Quality (IEQ) focuses on occupant comfort through low-VOC materials and adequate ventilation.

**Answer framework:** Compare the GRIHA and LEED rating systems based on their assessment criteria; explain the importance of 'Indoor Air Quality' (IAQ) for occupant health.

**Important point:** Rating systems are tools for assessment, not substitutes for building codes. Certification must be performed by authorised third-party agencies based on verified project data.

PRACTICE ONLY

# Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

### Part A — short response

1. State one key definition from Module 1: Green Building Fundamentals and Site Planning.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Energy Efficient Design and Passive Strategies.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Green Materials and Water Conservation.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Energy Codes, Auditing and Rating Systems.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

### Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

### Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and characteristics of green buildings; benefits (environmental, economic, social); sustainable site selection and development; minimizing site disturbance; heat island effect; ecological footprint of buildings..
2. Develop a complete answer or practical plan for: Building orientation and form; shading devices (internal and external); thermal insulation; building envelope design; passive heating and cooling techniques (ventilation, thermal mass); energy efficient lighting and HVAC..
3. Develop a complete answer or practical plan for: Sustainable building materials (low-carbon cement, recycled aggregate, fly ash); embodied energy of materials; water conservation strategies; rainwater harvesting; greywater treatment and reuse; low-flow fixtures..
4. Develop a complete answer or practical plan for: Energy Conservation Building Code (ECBC); principles of energy auditing; green building rating systems in India (GRIHA, LEED, IGBC); indoor environmental quality (VOC, ventilation); life cycle cost analysis..

LAST-MILE REVISION

## Quick Revision

### Module 1: Green Building Fundamentals and Site Planning

Definition and characteristics of green buildings; benefits (environmental, economic, social); sustainable site selection and development; minimizing site disturbance; heat island effect; ecological footprint of buildings.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Energy Efficient Design and Passive Strategies

Building orientation and form; shading devices (internal and external); thermal insulation; building envelope design; passive heating and cooling techniques (ventilation, thermal mass); energy efficient lighting and HVAC.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Green Materials and Water Conservation

Sustainable building materials (low-carbon cement, recycled aggregate, fly ash); embodied energy of materials; water conservation strategies; rainwater harvesting; greywater treatment and reuse; low-flow fixtures.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Energy Codes, Auditing and Rating Systems

Energy Conservation Building Code (ECBC); principles of energy auditing; green building rating systems in India (GRIHA, LEED, IGBC); indoor environmental quality (VOC, ventilation); life cycle cost analysis.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

## Glossary

| Important terms from the syllabus | |
|---|---|
| **Term / topic** | **One-line meaning** |
| Green building concepts and benefits | Green buildings aim to reduce the overall impact of the built environment on human health and the natural environment by efficiently using energy, water and other resources. |
| Sustainable site selection and design | Site planning involves selecting locations with good connectivity, protecting existing vegetation and minimizing soil erosion. |
| Orientation and shading | Proper orientation maximizes natural light and minimizes unwanted solar heat gain. |
| Passive cooling and heating | Passive strategies use natural processes to maintain comfort. |
| Sustainable materials and embodied energy | Green materials have low 'embodied energy'—the energy required for extraction, processing and transport. |
| Water conservation and management | Buildings conserve water through demand reduction (low-flow fixtures), rainwater harvesting for recharge or use, and onsite treatment and reuse of greywater for landscaping and flushing, significantly reducing the load on municipal supplies. |
| Energy codes and auditing | The ECBC sets minimum energy performance standards for commercial buildings. |
| Green rating systems and IEQ | Rating systems like GRIHA (National) and LEED/IGBC (International) provide a framework for assessing and certifying building sustainability. |

LEARNING RESOURCES

# References

### Recommended green-building references


1. K. S. Jagadish, Sustainable Building Technology.
2. GRIHA Manuals (Vol. I to V).
3. Bureau of Energy Efficiency (BEE), Energy Conservation Building Code (ECBC).
4. S. M. Patil, Green Building.


### Approved public green-building sources

- https://nptel.ac.in/courses/105102195
- https://onlinecourses.swayam2.ac.in/nou24_ge85/preview

These sources provide the verified study basis while official Course 6014C detail is unavailable; they do not replace official energy audits, authorised green-building certifications, certified material testing or legal building codes.